

Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 7: Weather, Climate and Adaptations of Animals to Climate

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	7 — Weather, Climate and Adaptations of Animals to Climate
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps: reading one day's weather as proof of climate (or the reverse); confusing the value of an extreme with the time it occurred; pairing a feature with an incidental side effect instead of its real survival job; and applying surface-area-to-volume logic the wrong way round for hot versus cold animals. Do not write on this paper — mark answers on your answer sheet.

1. A researcher has 35 years of daily records for a town. She computes, for each calendar month, the long-run average maximum temperature and average rainfall, then publishes a single chart titled 'what this town is generally like through the year'. The single most important reason her chart describes climate and not weather is that:

- (A) it includes both temperature and rainfall, and using two elements at once turns weather into climate
- (B) each value on it is an average built from many years of data, so a single unusual day cannot move it
- (C) it is drawn as a chart rather than written as a sentence, and charts always show climate
- (D) it covers a whole year, and any record spanning twelve months automatically counts as climate

2. Town P's yearly rainfall over five years was 1980, 2050, 1990, 2110 and 2020 mm, falling in almost every month. Town Q's was 210, 190, 230, 180 and 200 mm, nearly all in

two short months. Even though you have only five years, the strongest climate conclusion is that:

- (A) the two towns have the same climate because both show some variation from year to year
- (B) neither town can be assigned a climate, because five years is far too short to mean anything
- (C) Q has the wetter climate, because its rainfall is more concentrated and therefore more intense
- (D) P has a consistently wet, evenly-spread climate and Q a dry, strongly-seasonal one, because each town's pattern repeats year after year

3. A travel blog claims: 'I visited the Thar region in the one week it rained, so its climate is actually wet.' A geography student replies that the blog has confused two ideas. The precise error is that the blogger:

- (A) forgot that desert climates change from one day to the next, so any week could be wet
- (B) treated a brief spell of unusual weather as if it defined the long-term climate of the region
- (C) assumed rainfall is never a part of weather, when in fact it is only a part of climate
- (D) believed deserts can never receive any rain at all, which is what made the trip misleading

4. A city's record shows that its hottest-ever day and its coldest-ever day both occurred within the last 40 years, and yet its 40-year average temperature has barely changed. The best interpretation is that:

- (A) the city's climate must be unstable, because it contains both record heat and record cold
- (B) the record days must be measurement errors, since a stable average rules out true extremes
- (C) the hot and cold records cancel out exactly, which is the only way an average can stay fixed
- (D) extreme single days are part of normal weather variation and do not by themselves shift the long-term climate average

5. Which of these questions could in principle be answered correctly from a single day's weather report, and which needs many years of records? Consider: (i) 'Is it raining outside right now?' (ii) 'Which month here is usually the driest?' The correct pairing is:

- (A) (i) needs only today's report; (ii) needs many years of climate data
- (B) both can be answered from a single day's report, since both mention rain or dryness

(C) both need many years of data, because any question about a place is about its climate

(D) (i) needs many years of data and (ii) needs only today's report, since 'right now' implies a long history

6. Two friends each describe their home town. Asha: 'My town gets cold winters and a heavy monsoon every year.' Bina: 'It is drizzling in my town this afternoon.' Which describes climate, which describes weather, and can both be true of the same town?

(A) Asha describes climate and Bina describes weather, and both could be true of one town at once

(B) both describe weather, because each statement mentions rain or temperature

(C) Asha describes weather and Bina describes climate, because Asha's is more detailed

(D) they cannot both be true of the same town, since one mentions rain and the other does not

7. A student reasons: 'Yesterday was 12 °C and today is 27 °C in the same city, a 15-degree jump, so the climate must be shifting fast.' The flaw is that:

(A) a one-day temperature swing is ordinary weather variation, which does not by itself indicate any change in climate

(B) a 15-degree change is impossible in one day, so the readings must be wrong

(C) the student is right, because two consecutive different days prove the climate is changing

(D) temperature cannot be used to judge climate at all, only rainfall can

8. A planner must decide (i) what to wear on a walk in the next hour and (ii) in which months to schedule the town's yearly flood-defence repairs. The correct match of each decision to weather or climate is:

(A) both should be based on the town's climate, since planning is always long-term

(B) both should be based on today's forecast, since both are decisions made today

(C) (i) should use today's weather forecast; (ii) should use the town's long-term climate

(D) (i) should use climate and (ii) should use the daily forecast, since flood timing changes hourly

9. A six-year set of average annual temperatures for a city reads 26.0, 26.1, 25.9, 26.2, 26.0, 26.1 °C, while a second city reads 14, 22, 9, 25, 11, 23 °C for the same six years. Comparing the two, the soundest statement about their CLIMATES is:

(A) both cities have the same climate, because their six-year average temperatures come out close

(B) the second city has no climate, because a place whose yearly figures jump around cannot have one

(C) the first city has no weather, because temperatures that barely change mean nothing is happening

(D) the first city has a steady year-to-year climate while the second swings widely, a contrast that holds across the whole six-year run

10. A weather bulletin lists: temperature 31 °C, humidity 64%, rainfall 6 mm, wind 22 km/h, sunshine 5 hours. A teacher asks which TWO of these figures could NOT be obtained from the same single instrument as the wind speed. The correct pair (excluding the wind figure itself) is:

(A) the temperature of 31 °C and the sunshine of 5 hours, because both depend directly on the Sun

(B) the humidity of 64% and the temperature of 31 °C, because both are measured by the same hygrometer

(C) the rainfall of 6 mm and the humidity of 64%, since each needs its own dedicated instrument different from the anemometer

(D) the rainfall of 6 mm and the sunshine of 5 hours, because both are forms of falling matter

11. Rainfall is reported as a depth in millimetres rather than as the number of litres a gauge collects. The deepest reason this makes data from different stations comparable is that:

(A) litres cannot be measured by any rain gauge, so depth is the only quantity left to report

(B) depth does not depend on the width of the gauge's mouth, so the same rainstorm gives the same depth on a narrow gauge and a wide one

(C) depth measures how fast the rain fell, and speed is what matters for comparing storms

(D) a wide gauge always collects fewer litres than a narrow one, which would make litres misleading

12. A maximum-minimum thermometer is read once each evening. On a particular day it shows max 33 °C and min 18 °C, but the observer was away all afternoon and watched nothing. The feature of this instrument that makes the readings trustworthy is that it:

(A) retains a record of the highest and lowest temperatures reached during the day even when no one is watching

(B) continuously displays the exact temperature at the precise moment it is read, which is enough

(C) measures both the temperature and the rainfall over the day in one combined reading

(D) averages all the temperatures of the day automatically to give the max and the min

13. A five-day record reads — Max (°C): Mon 30, Tue 27, Wed 36, Thu 34, Fri 31; Min (°C): Mon 19, Tue 20, Wed 17, Thu 26, Fri 21. The day with the LARGEST daily temperature

range was:

- (A) Thursday, because it had the highest minimum temperature of the week
- (B) Tuesday, because it had the lowest maximum temperature of the week
- (C) Wednesday, with a range of 19 °C
- (D) Monday, because both its readings were the lowest pair early in the week

14. Using the same table (Max: Mon 30, Tue 27, Wed 36, Thu 34, Fri 31; Min: Mon 19, Tue 20, Wed 17, Thu 26, Fri 21), which single statement is fully correct?

- (A) Thursday had the coolest night, because its minimum number 26 looks like a night reading
- (B) Wednesday had both the hottest day and the coolest night of the week
- (C) Tuesday had the hottest day, because the week's heat built up to a peak by mid-week
- (D) Monday had the coolest night, because the earliest day of the week is usually the coldest

15. A week's rainfall (mm) is: Mon 0, Tue 9, Wed 0, Thu 0, Fri 15, Sat 30, Sun 6. A report must give the number of rainy days, the weekly total, and the average rain PER RAINY DAY. The correct set is:

- (A) 4 rainy days, total 60 mm, average 15 mm per rainy day
- (B) 7 rainy days, total 60 mm, average about 8.6 mm per day
- (C) 4 rainy days, total 60 mm, average about 8.6 mm per rainy day
- (D) 3 rainy days, total 54 mm, average 18 mm per rainy day

16. A humidity log for one day reads: 05:00 → 90%, 09:00 → 78%, 13:00 → 55%, 17:00 → 64%, 21:00 → 86%. The reading dips to its lowest at 13:00 chiefly because:

- (A) the air actually held the least water vapour of the whole day at exactly 13:00 because vapour escapes at noon
- (B) early-afternoon sunshine has warmed the air to its hottest, and warmer air's relative humidity reads lower for the same amount of moisture
- (C) the hygrometer always resets itself to a low value in the early afternoon
- (D) the wind was strongest at 13:00 and blew the moisture out of the instrument

17. A rain gauge has a collecting jar 150 mm deep. After an exceptional storm the jar is brimful and overflowing, and the observer writes '150 mm'. The key lesson for reliable weather data is that:

- (A) the rainfall was exactly 150 mm, since that is the figure the full gauge shows
- (B) the gauge has started measuring wind once it overflowed, so the figure is no longer rainfall

(C) the true rainfall was at least 150 mm; an overflowing gauge can only set a lower bound, not the exact amount

(D) the overflow means no further rain fell after the jar filled, so 150 mm is the final total

18. Over four days a city's maximum temperatures were 28, 30, 35 and 31 °C and its minimum temperatures were 18, 20, 21 and 17 °C. Which figure is closest to the AVERAGE DAILY RANGE over the four days?

(A) 12 °C

(B) 18 °C, the average of the four maximum temperatures minus the lowest minimum

(C) 31 °C, the average of the four maximum temperatures

(D) 47 °C, the highest maximum minus the lowest minimum across all days

19. A station logs hourly temperatures and finds the air is hottest at about 2 p.m., even though the Sun is highest at noon. A graph of sunlight intensity peaks at noon while the temperature graph peaks two hours later. This LAG happens because:

(A) the Sun is physically closest to the Earth at 2 p.m. rather than at noon

(B) the thermometer reacts slowly and takes two hours to catch up with the real temperature

(C) humidity reaches zero at 2 p.m., and dry air is what makes the afternoon hottest

(D) the ground keeps absorbing more heat than it loses for a while after the Sun's peak, so temperature continues rising past noon

20. The Sun is called the ultimate driver of weather. Which chain of cause and effect best captures HOW its energy ends up producing wind and rain?

(A) the Sun's rays physically push against the air with their pressure, shoving it sideways to make wind

(B) the Sun cools the oceans so that the cooled water can rise as vapour and fall again as rain

(C) moonlight at night and sunlight by day share the work equally, the Moon driving the wind and the Sun the rain

(D) sunlight heats land and sea unevenly, the warmed air rises and creates pressure differences that drive wind, while heat also evaporates water that later condenses into clouds and rain

21. Places near the equator are warmer year-round than places near the poles. The correct Sun-based reason, and the wrong popular one, are:

(A) near the equator the Sun's rays strike more directly, concentrating their energy on a smaller patch of ground, whereas near the poles the same beam is spread thin over a larger area

(B) the equator is thousands of kilometres physically closer to the Sun than the poles are

- (C) the Sun shines for a full 24 hours every day at the equator but never at the poles
- (D) the equator simply has more cities and machines, and these are what keep it warm

22. During a polar winter the Sun may stay below the horizon for weeks, while in polar summer it may not set for weeks. The most accurate statement linking this to the region's CLIMATE is:

- (A) the weeks of midnight Sun in summer make the polar climate hot for half the year
- (B) the missing Sun in winter makes the region suddenly hot, since darkness traps heat
- (C) the long days and long nights cancel out, giving the poles a mild, temperate climate
- (D) the long sunless winter keeps temperatures extremely low, and even the long summer days cannot fully warm the region, so its overall climate stays cold

23. On a clear afternoon a spot in full sunlight reads 38 °C while a shaded spot a few metres away, at the same time on the same clock, reads 30 °C. The clearest thing this single comparison demonstrates is that:

- (A) the clock controls temperature, since both readings were taken at the same time
- (B) it is incoming sunlight, not merely the time on the clock, that does the heating, since equal-time spots differ by where the sunlight falls
- (C) shade is colder because cold air sinks only into shaded places
- (D) the thermometer in the shade must be faulty, since two thermometers should always agree

24. A meteorologist can forecast tomorrow's weather fairly well but cannot say what the weather will be on this exact date next year, though she CAN describe next year's likely season. The reason this is consistent, not contradictory, is that:

- (A) the Sun grows too weak a year ahead to make any weather, so only seasons remain knowable
- (B) climate is impossible to study, so only short-range weather can ever be predicted
- (C) thermometers stop working after a single day, which is why long-range daily forecasts fail
- (D) small differences in the many interacting atmospheric factors grow rapidly, blurring far-off daily detail, while the averaged seasonal pattern (climate) remains predictable

25. Why is the SAME slanting-rays principle that makes the poles cold also the reason a winter noon feels weaker than a summer noon at one mid-latitude city?

- (A) in winter the Sun moves much farther from the Earth, weakening every ray it sends
- (B) in winter the Sun stays lower in the sky, so its rays strike more slantingly and spread their energy over more ground, just as polar rays do
- (C) in winter the Sun actually shines for more hours but each hour carries less heat
- (D) winter air is thicker and physically blocks most of the sunlight from reaching the ground

26. A region near the equator and a region near a pole receive sunlight. Over a full year, which statement about the SUN'S ENERGY reaching each correctly explains their climates?

- (A) both regions receive the same energy per unit area, so any climate difference must come from the wind alone
- (B) the poles receive more total sunlight because their summer days are very long, so they should be warmer
- (C) the equator is cold and the poles are hot, because slanting rays carry more heat than direct ones
- (D) the equator receives intense, nearly overhead sunlight all year, giving a hot climate, while the poles receive weak, slanting sunlight, giving a cold climate

27. Four cities report (warmest-month average, coolest-month average, yearly rainfall): City A (29 °C, 27 °C, 2300 mm); City B (45 °C, 8 °C, 120 mm); City C (4 °C, -28 °C, 180 mm); City D (33 °C, 18 °C, 700 mm). Which city's data most clearly fits a TROPICAL RAINFOREST climate?

- (A) City B, because its warmest month is the hottest of the four
- (B) City A, because it is consistently hot with a tiny seasonal swing and very heavy rainfall
- (C) City C, because its rainfall and one of its temperatures are similar to City B's
- (D) City D, because it is warm and has moderate rainfall typical of forests

28. A data card describes a habitat as: temperature 24-29 °C every month, humidity above 85%, rainfall over 2200 mm spread through the year, dense tall canopy. An animal with strong grasping hands, a long tail and loud calls fits here. The single best joined-up explanation is:

- (A) the constant heat, heavy rain and tall trees define a rainforest, where grasping limbs aid climbing and loud calls carry through dense foliage that blocks sight
- (B) the figures describe a polar region, where grasping tails grip the ice and loud calls echo across open snow
- (C) the figures describe a hot desert, where a long tail stores water and loud calls travel far over sand
- (D) the figures describe a cold mountain snowfield, where strong limbs dig through snow and calls warn of cold

29. A nature note states: 'A thick fat layer, a compact rounded body and white fur all serve one common climatic purpose.' What is that single shared purpose, and what would happen to such an animal in a tropical rainforest?

- (A) all three help the animal lose heat, so it would stay comfortably cool in the rainforest
- (B) all three conserve heat or hide the animal in snow, so in a hot wet rainforest it would overheat and be conspicuous against green leaves

(C) all three are for camouflage among green leaves, so it would be perfectly hidden in the rainforest

(D) all three store water, so it would survive the rainforest easily by drawing on its reserves

30. Two penguins differ only in body shape: Penguin X is compact and rounded; Penguin Y is unusually long and slender with a large thin tail. In the same freezing Antarctic conditions, the likely outcome is that:

(A) Penguin Y survives better because its larger surface area soaks up more of the Sun's warmth

(B) both survive equally, because body shape has no effect on how an animal loses heat

(C) Penguin X loses heat faster because a compact body has more surface exposed to the cold

(D) Penguin X conserves heat better because its lower surface-area-to-volume ratio loses heat more slowly, while Penguin Y loses heat too fast

31. A polar bear's coat is dense and water-shedding, sitting over a thick fat layer; the bear also has wide paws and is a strong swimmer. Which statement pairs EACH feature with its correct main job?

(A) the fat layer's main job is to make the bear float higher than its prey so it can ambush from above

(B) the dense fur's main job is to keep the bear cool through the brief polar summer

(C) the wide paws spread weight on snow and act as paddles in water, while the fur-plus-fat combination slows heat loss in air and icy water

(D) the swimming ability's main job is to let the bear climb trees when the ice melts

32. A reviewer writes: 'The polar bear is white so that it can absorb the most sunlight and keep warm.' The correct response is that the white coat is:

(A) correct, because white surfaces absorb the most heat from sunlight

(B) mainly camouflage that blends the bear with snow and ice while it hunts, not a device for absorbing the Sun's heat

(C) white only so other bears can spot it easily across the ice

(D) white so that its prey can see it coming and the hunt is fair

33. Penguins press into a tight huddle during the coldest, windiest spells. The correct cause-and-effect account is that huddling:

(A) increases each bird's exposed surface, helping the group shed dangerous excess heat

(B) lets the birds catch fish more easily because crowding herds the fish toward them

(C) reduces the total body surface each bird exposes to the cold air, so the group as a whole loses heat more slowly

(D) traps a pocket of cold air between the birds, which cools them and saves energy

34. An elephant in a hot tropical region has very large, thin ears richly supplied with blood vessels, which it flaps. A polar fox has small, well-furred ears. The single principle that explains BOTH ear designs is:

- (A) both kinds of ear are mainly about hearing, the large ones hearing distant sounds and the small ones hearing under snow
- (B) large ears trap heat to keep the elephant warm, and small ears trap heat to keep the fox warm
- (C) ear size is set by body size alone and has no connection to the climate the animal lives in
- (D) large thin ears expose much surface to shed excess heat in the warmth, while small ears expose little surface to conserve heat in the cold

35. Many rainforest monkeys and frogs are bright green and stay among leaves, while polar bears are white against snow. The deep idea common to both colourings is that an animal's colour is favoured when it:

- (A) is the colour that best warms its body, green absorbing heat in forests and white in snow
- (B) glows in the dark, helping it to be seen by its own kind in any habitat
- (C) is the brightest possible, so that predators are frightened away by the bold colour
- (D) matches the usual background of its habitat, hiding it from predators or prey in that particular climate's surroundings

36. A monkey adapted with grasping hands, a long balancing tail and loud calls is moved to a polar ice region. It would most likely struggle because every one of these features is suited to:

- (A) moving and communicating in a warm, wet, densely-treed forest, none of which exists on open polar ice
- (B) surviving intense cold, so the only problem would be the unfamiliar terrain
- (C) burrowing in hot dry sand, which is why the ice would defeat it
- (D) flying long distances over open ocean, which the ice prevents

37. Why do tropical rainforests support a far greater VARIETY of animals than polar regions do? The best causal explanation is that:

- (A) polar cold makes animals reproduce much faster, but the rainforest still wins because it is bigger
- (B) rainforests have almost no plants, so animals there face no competition and multiply freely
- (C) the warm, wet, stable rainforest climate provides abundant and varied food and shelter all year, allowing many different kinds of animal to coexist
- (D) variety depends mainly on how many human visitors a region receives each year

38. A toucan has a very large but light beak; a hummingbird a slender one; a polar bear thick fur. A student must match each to the right reason. Which matching is fully correct?

- (A) the toucan's beak stores body fat for a cold season, while the bear's fur keeps it cool in summer
- (B) the toucan's beak digs burrows in dry soil, while the bear's fur camouflages it among leaves
- (C) the toucan's light beak reaches fruit on thin branches it could not perch on, while the bear's thick fur insulates it against cold
- (D) the toucan's beak warms its head on freezing nights, while the bear's fur helps it swim faster

39. A habitat table lists: Habitat M (average 27 °C, 2400 mm rain, dense canopy) and Habitat N (average -22 °C, 160 mm precipitation, ice fields). An animal with two fur layers, a fat store, a compact body and white colour belongs in:

- (A) Habitat M, because thick fur is most needed where rain is heaviest
- (B) Habitat M, because white colour blends best with green leaves
- (C) Habitat N, whose freezing, icy conditions match its heat-conserving, snow-camouflaged features
- (D) either habitat equally, because fur insulates the same in any climate

40. A red-eyed tree frog sits motionless and green on a leaf by day and is hard to spot. The strongest evidence that its colour is a genuine ADAPTATION, rather than a lucky coincidence, is that:

- (A) the colour keeps the frog warm during cold nights, which is useful in any climate
- (B) the colour lets the frog reflect sunlight and stay cool, as a desert animal would need
- (C) the colour allows the frog to glow and lure prey toward it in complete darkness
- (D) the colour reliably matches the green leaves of its rainforest home, reducing the chance predators detect it, generation after generation

41. An animal expert defines an adaptation as 'a heritable feature that, over long times, suits an animal to its climate.' Which observation best fits this definition?

- (A) a single bear that happened to feel warm on one unusually mild afternoon
- (B) a monkey that learns to open a new kind of fruit during one morning
- (C) a bird that flies indoors to escape a cold night on one occasion
- (D) polar bears born generation after generation with thick fur and a fat layer, matching the long-cold polar climate

42. A Siberian crane breeds in northern Russia and is seen at Indian wetlands only in winter, vanishing each Indian summer. Which statement correctly distinguishes the ROLE OF CLIMATE from the role of day-to-day weather in this migration?

- (A) the crane reacts to the particular weather of the one day it happens to set out, choosing only sunny days
- (B) the crane follows the regular seasonal climate cycle, leaving the harsh northern winter and returning when the north warms, not responding to any single day's weather

- (C) the crane dies out each Indian summer and a fresh set appears each winter, driven by the day's conditions
- (D) the crane changes into a heat-loving species for the Indian summer, a change triggered by hot weather

43. A six-day sunshine record (hours) reads: Mon 9, Tue 1, Wed 8, Thu 0, Fri 7, Sat 2. A student must state what this best illustrates. The soundest reading is:

- (A) the figures prove the region's climate changed during the single week
- (B) sunshine swung sharply from day to day, illustrating how quickly weather can change over a short period
- (C) sunshine was effectively constant, illustrating a stable climate
- (D) the Sun shone for the same hours each day regardless of the clouds present

44. A weather summary reads: 'Maximum 5 °C, minimum -15 °C, light snow, 19 hours of darkness.' A second reads: 'Maximum 30 °C, minimum 26 °C, heavy rain, dense forest nearby.' Matching each summary to a climate type, the correct pairing is:

- (A) the first points to a hot desert because its temperature range is wide, and the second to a polar climate because it has precipitation
- (B) the first points to a polar-type climate and the second to a tropical-rainforest climate
- (C) both point to tropical climates because both report some form of precipitation
- (D) the first points to a temperate coast because its maximum is above zero, and the second to a desert because it is hot

45. A child argues: 'It is freezing in our town today, so we must be living in a polar region.' The clearest reason this is mistaken is that:

- (A) cold days simply never occur anywhere except in polar regions
- (B) a single cold day really does prove a polar climate, so the child is right
- (C) one freezing day is weather and can happen in many places, while a polar region is defined by cold persisting through most of the year
- (D) temperature plays no part in defining whether a region is polar

46. Two stations at the SAME latitude are compared over a year. Coastal station C has a small annual temperature range and high humidity; inland station I has a large annual range and lower humidity. Since latitude is equal, the best explanation for the contrast is that:

- (A) station C must lie at a much higher altitude, which is what shrinks its temperature range
- (B) the nearby sea moderates station C's temperatures and adds moisture, while inland station I, far from the sea, swings more widely and stays drier

(C) station I receives more direct sunlight than C because inland places are closer to the Sun

(D) the two differ only in the single days on which they were measured, not in any lasting way

47. A table compares the day-to-day MAXIMUM temperature of two places over six days. Place 1: 27, 28, 26, 28, 27, 28 °C. Place 2: 42, 19, 44, 17, 41, 18 °C. The averages happen to be similar. The most accurate climatic statement is:

(A) Place 1 has small day-to-day swings and Place 2 has very large ones, a difference in their climates that the similar averages hide

(B) the two places have identical climates, because their six-day average maxima come out close

(C) Place 2 has no climate at all, because a place whose temperature jumps so much cannot have one

(D) Place 1's steadiness proves it has no weather, since nothing about it changes

48. A weather page reports for one city: temperature 33 °C, humidity 70%, rainfall 0 mm, wind 25 km/h, sunshine 7 hours. A reader wants the ONE figure that tells how much water vapour the air holds. The correct choice, and why the others mislead, is:

(A) the rainfall of 0 mm, because rainfall directly counts the water in the air

(B) the temperature of 33 °C, because hotter air always means more vapour is present

(C) the humidity of 70%, because humidity is the measure of water vapour in the air, given as a percentage

(D) the wind speed of 25 km/h, because wind carries the moisture and so measures it

49. An animal is described only as having 'a long, slender body with large, thin, well-exposed ears and short fur'. Without naming a species, the climate this body plan is best suited to, and the reason, is:

(A) a polar climate, because large ears catch and hold warmth against the cold

(B) a polar climate, because a slender body conserves heat better than a compact one

(C) a hot climate, because the large exposed surface and thin fur help the animal shed body heat rather than trap it

(D) any climate equally, because body shape and ear size do not affect heat loss

50. A scientist plots monthly average temperature for one city: it rises smoothly from a January low to a July high, then falls back, repeating the same curve each year for ten years. A second city's daily plot for one week zig-zags up and down with no repeating shape. The first plot shows _____ and the second shows _____:

(A) the repeating yearly curve shows climate, while the zig-zag week shows weather

(B) both plots show climate, because both are drawn from measured temperatures

(C) the yearly curve shows weather and the weekly zig-zag shows climate, because the longer plot is more detailed

(D) both plots show weather, because temperature is always a weather element

51. A flock breeds at a northern wetland each spring and flies south each autumn, returning north the next spring, repeating this for many years. What governs this regular back-and-forth, and what early cue most likely starts the autumn departure?

- (A) the random weather of a single chosen day governs it, and a sudden cold snap on that day is the cue
- (B) the birds permanently settling governs it, and the cue is that they simply stop wanting to fly
- (C) the regular seasonal climate cycle governs it, and shortening day length together with a falling food supply is the most likely cue that starts the southward flight before the worst cold arrives
- (D) a one-time event that never repeats governs it, and the cue is the colour of one tree's leaves

52. A rain gauge at Station A has a wide funnel; an identical-depth gauge at Station B has a narrow funnel. The same rainstorm passes over both. What will their readings show, and what does this reveal about why depth is used?

- (A) both will read the same depth, because depth is independent of funnel width, which is exactly why depth lets stations be compared fairly
- (B) the wide gauge will read a greater depth, because it collects more water overall
- (C) the narrow gauge will read a greater depth, because the same water is squeezed into a thinner column
- (D) the two cannot be compared at all, because their funnels differ in size

53. A region records, every year, summer maxima around 47 °C, winter minima around 5 °C, almost no rain, and very wide day-to-night temperature swings. Which combined reading of these climate signatures is correct?

- (A) this is a tropical rainforest, because the high summer maxima show it is hot and wet
- (B) this is a polar climate, because the wide temperature swings resemble polar extremes
- (C) this is a temperate coastal climate, because the winter minimum stays above freezing
- (D) this is a hot-desert climate, where clear dry air heats strongly by day and loses heat fast at night, giving large swings and little rain

54. Two adaptations are: (i) a thick layer of fat under the skin and (ii) long limbs with a grasping tail. Each is an advantage in one climate and a disadvantage in the other. Which statement correctly assigns both?

- (A) both features help in the cold, the fat for warmth and the grasping tail for gripping ice
- (B) the fat layer aids a cold-climate animal but would cause overheating in a hot rainforest, while grasping limbs aid a rainforest climber but are useless on bare polar ice

(C) both features help in the rainforest, the fat for cushioning against rain and the tail for climbing

(D) neither feature depends on climate, since fat and limbs serve the same purpose everywhere

55. Why can a meteorologist say 'next July will probably be the wettest month here' yet be unable to say whether it will rain on the 14th of next July? The single principle that resolves this apparent puzzle is:

(A) the long-term averaged climate (which month is usually wettest) is predictable, but the exact weather of one distant day is not, because tiny atmospheric differences grow rapidly over time

(B) the Sun behaves predictably in July but unpredictably on individual days, so only months can be forecast

(C) rainfall obeys the calendar, so months are fixed but individual dates are chosen at random by the weather

(D) thermometers and rain gauges work for whole months but fail when asked about a single future day

56. A naturalist notes that a polar animal has a compact rounded body while a similar-sized desert animal of a related kind is long and lean with big ears. Asked to state the ONE underlying rule, the best answer is:

(A) surface-area-to-volume ratio governs heat exchange: a compact body has less surface for its volume and conserves heat in the cold, while a lean body and big ears have more surface and shed heat in the warmth

(B) compact bodies lose heat faster, which is what the cold demands, while lean bodies lose heat slower, which suits the heat

(C) body shape only changes how large each animal looks to predators and has nothing to do with heat

(D) the desert animal is lean because food is scarce, and the polar animal is round because food is plentiful

57. A student combines three facts: (1) the Sun heats the equator more strongly than the poles; (2) warm air rises and cool air sinks; (3) rainforests cluster near the equator while ice caps lie at the poles. The best single conclusion tying these together is:

(A) stronger, more direct equatorial sunlight makes that zone hot and supports wet rainforest climates, while weak slanting polar sunlight leaves those zones cold and icy

(B) the poles are hot because warm air sinks there, which is why ice forms only at the equator

(C) the equator is cold because rising warm air leaves it empty of heat, so rainforests need the cold

(D) sunlight strength has no link to where rainforests and ice caps occur; their positions are accidental

58. A weather observer records temperature every hour and notices it falls steadily through the night, reaching its LOWEST value just before sunrise rather than at midnight. The best Sun-based explanation is that:

- (A) midnight is the moment the Earth is farthest from the Sun each day, so it should be coldest then
- (B) the Moon actively pulls heat from the ground until midnight, after which the air warms again
- (C) the thermometer drifts lower overnight on its own, regardless of how much heat the ground has lost
- (D) the ground keeps losing its stored heat all night with no incoming sunlight, so the air stays coldest after the longest period of cooling, just before the Sun returns

59. A graph shows daytime air temperature rising from sunrise, peaking in the early afternoon, then falling toward sunset, while a second curve on the same axes shows the Sun's height in the sky peaking exactly at noon. The fact that the temperature peak lags BEHIND the noon Sun peak is best explained by reasoning that:

- (A) the Sun delivers its greatest heat in mid-afternoon because it is then directly overhead
- (B) the air takes a fixed two-hour time to physically travel from the Sun down to the ground
- (C) even after the Sun begins to descend from noon, the ground is still gaining heat faster than it loses it for a couple of hours, so the air keeps warming past noon
- (D) clouds always gather at noon and block the Sun, so the heating is delayed until they clear

60. A final synthesis question. Which single statement is fully correct about weather, climate and adaptation together?

- (A) climate is the short-term state and weather is the many-year average, and adaptations respond to each day's weather
- (B) weather and climate are the same measurement, and adaptations are habits an animal learns within a single day
- (C) weather is the short-term state of the atmosphere at a place, climate is the average of that weather over many years, and animals' adaptations are heritable features matching the long-term climate, not the weather of one day
- (D) weather is measured by instruments but climate cannot be measured in any way, and adaptations have nothing to do with climate

Solutions — Science (Class 7), Chapter 7: Weather, Climate and Adaptations of Animals to Climate — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	B	21	A	31	C	41	D	51	C
2	D	12	A	22	D	32	B	42	B	52	A
3	B	13	C	23	B	33	C	43	B	53	D
4	D	14	B	24	D	34	D	44	B	54	B
5	A	15	A	25	B	35	D	45	C	55	A
6	A	16	B	26	D	36	A	46	B	56	A
7	A	17	C	27	B	37	C	47	A	57	A
8	C	18	A	28	A	38	C	48	C	58	D
9	D	19	D	29	B	39	C	49	C	59	C
10	C	20	D	30	D	40	D	50	A	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) Climate is the average of weather over a long span of years; the defining feature here is the multi-year averaging that smooths out individual days. Using two elements does not change weather into climate — a single day can list both temperature and rainfall and still be weather. The format of the output (chart versus sentence) is irrelevant to whether something is climate. Covering one year is not enough either; a year of daily readings is still a year of weather, and climate needs averaging across many years.

2. (D) Climate is a repeating long-term pattern. P's totals are large and steady with rain spread through the year (wet, even); Q's are small and squeezed into two months (dry, seasonal) — and crucially both patterns recur across all five years, which is what lets you read climate from them. Slight year-to-year wobble does not make their climates the same; the totals differ roughly tenfold. Five years already shows a clear repeating signal here.

Concentrating a small total into two months does not make Q wetter — Q receives far less rain overall.

3. (B) The blogger generalised one wet week (weather) into a claim about climate, the long-term average — the central weather-versus-climate trap. Climate does not flip day to day; that is precisely the property the blogger ignored. Rainfall is an element of weather (and contributes to climate too), so it is wrong to say it is never weather. Deserts do occasionally receive rain, so the error is not the belief that they never do — it is reading one wet week as the climate.

4. (D) A long average is robust: individual extreme days, hot or cold, are weather and barely move a 40-year mean. The presence of extremes does not make the climate unstable — extremes are expected within a stable climate. There is no reason to call the records errors; real extremes coexist with a steady average. And the average is not held fixed by extremes 'cancelling' — it stays steady because each extreme is one day among roughly fourteen thousand, far too small a share to shift the mean.

5. (A) Whether it is raining now is the instantaneous state of the atmosphere — pure weather, answerable from one report. 'Which month is usually driest' asks about a typical, averaged pattern — climate, needing many years. Both cannot come from one day, because 'usually driest' is an average. Both do not need long records either, because the 'right now' question is settled by current weather. The reversed pairing is wrong because 'right now' is the very thing a single report tells you.

6. (A) Asha gives a recurring yearly pattern (climate); Bina gives the present state of the air (weather). Mentioning rain or temperature does not make a statement weather — a climate description also uses those elements. The labels are not assigned by length or detail, so Asha's longer statement is not therefore weather. And the two are entirely compatible: a town with a wet-and-cold-winter climate can certainly be drizzling on a given afternoon.

7. (A) Day-to-day temperature changes, even large ones, are normal weather and say nothing about the long-term climate. A 15 °C swing between two days is perfectly possible, so the data need not be wrong. Two different days do not prove a climate shift — climate is the many-year average. Temperature is in fact one of the main elements used to describe climate, so it is not excluded from climate at all.

8. (C) Choosing clothes for the next hour depends on present conditions (weather); choosing which months to repair flood defences depends on which months usually flood (climate). Not both are climate — the next hour is governed by today's weather. Not both are today's forecast — repair scheduling spans a whole year and needs the averaged seasonal pattern. The reversed match fails because flood-prone months are a recurring climate feature, not something that changes hour by hour.

9. (D) Climate is judged from the pattern across years: the first city is remarkably steady, the second highly variable, and that contrast persists over all six years. Even if the two six-year means happened to be similar, the spread (stability versus large swings) is itself a climate difference, so 'same climate' is wrong. A place with variable yearly figures still has a climate — a variable one. And steady temperatures do not mean 'no weather'; weather still occurs daily even where the yearly average is constant.

10. (C) Wind speed is measured by an anemometer; rainfall needs a rain gauge and humidity needs a hygrometer — both distinct instruments, so they are the correct pair that the anemometer cannot give. Temperature and sunshine being Sun-related does not put them on one instrument; temperature uses a thermometer and they remain separate measurements. Humidity and temperature are not read by one hygrometer — a hygrometer measures humidity, not temperature. Sunshine is not 'falling matter' and rainfall and sunshine are measured quite differently, so that pairing's reasoning is wrong.

11. (B) For a given storm, a wider mouth catches proportionally more volume but over proportionally more area, so the DEPTH is the same — that station-independence is exactly why depth is reported. Rain gauges can measure volume; depth is preferred for comparability, not because volume is unmeasurable. Depth is an amount, not a rate, so it does not tell you how fast the rain fell. And a wider gauge collects MORE litres, not fewer, which is precisely why raw litres would not compare fairly between stations.

12. (A) A maximum-minimum thermometer mechanically marks the day's highest and lowest temperatures, so they can be read later — that is why the observer's absence does not matter. A reading of the current instant alone would miss the unobserved afternoon peak, so 'exact moment only' is wrong. It measures temperature, not rainfall, so it is not a combined instrument. And the max and min are not averages — they are the genuine extremes the instrument captured, not a mean of the day's readings.

13. (C) Daily range is max minus min: Mon 11, Tue 7, Wed 19, Thu 8, Fri 10 — Wednesday is largest at 19 °C. Thursday's high minimum (26 °C) actually shrinks its range to 8 °C, so a high minimum points the wrong way. Tuesday's low maximum gives only a 7 °C range, the smallest. Monday's range is 11 °C; having a low pair of readings does not by itself make a large gap between them.

14. (B) The highest maximum is Wed (36 °C) and the lowest minimum is also Wed (17 °C), so Wednesday holds both the hottest day and the coolest night. Thursday's minimum of 26 °C is the WARMEST night, not the coolest — a high minimum is a warm night. Tuesday's maximum of 27 °C is the lowest of the week, so it cannot be the hottest day. Monday's minimum is 19 °C, warmer than Wednesday's 17 °C, so Monday is not the coolest night; the order of the days in the week carries no temperature meaning.

15. (A) Days with rain are Tue, Fri, Sat, Sun = 4 days; total = 9 + 15 + 30 + 6 = 60 mm; average per rainy day = 60 / 4 = 15 mm. There are not 7 rainy days — three days had zero,

and dividing by 7 instead of 4 is the trap. Dividing the correct total by 7 gives about 8.6 and is the same trap with the right day-count omitted. And the total is 60, not 54, with 4 rainy days, not 3 — Tuesday's 9 mm must be counted.

16. (B) Relative humidity falls when air is warmed, and the air is hottest in the early afternoon thanks to accumulated solar heating — so the percentage dips even if moisture is unchanged. It is not that vapour 'escapes at noon'; the percentage can drop while the actual moisture stays similar, purely because warm air's capacity rises. Hygrometers do not auto-reset at a set time. Wind does not strip moisture out of a hygrometer; the dip is a temperature-driven relative-humidity effect.

17. (C) Once a gauge overflows it has hit its limit, so the recorded value is only a minimum — the real total could be much higher. It is therefore wrong to call 150 mm the exact amount. An overflowing rain gauge does not switch to measuring wind. And overflow does not mean rain stopped; rain may well have continued, with the extra simply spilling out unmeasured.

18. (A) Daily ranges are $28-18=10$, $30-20=10$, $35-21=14$, $31-17=14$; their average is $(10+10+14+14)/4 = 48/4 = 12$ °C. Subtracting one overall lowest minimum from an average maximum mixes incompatible quantities and gives a wrong 18. The average of the maxima alone (31) is not a range at all. Taking the single biggest max minus the single smallest min ($35-17=18$, or pairing extremes across days for 47) confuses the whole-period spread with the typical daily range.

19. (D) After noon the ground still gains heat faster than it sheds it, so the air keeps warming until gains and losses balance — that imbalance produces the afternoon lag. The Sun's distance does not change between noon and 2 p.m. in any way that matters; that idea is a misconception. A working thermometer responds quickly, so instrument lag is not the cause. And humidity does not fall to zero, nor is dryness the reason — the lag is about stored heat accumulating after the Sun's peak.

20. (D) Uneven solar heating sets up temperature and pressure differences (driving wind) and evaporates water (feeding clouds and rain) — a single energy source branching into both. Sunlight's physical push on air is far too weak to be the cause of wind; the cause is heating-driven pressure differences. The Sun warms, it does not cool, the oceans, and warming is what drives evaporation. The Moon does not supply the energy for weather; the Sun is the single ultimate source for both wind and rain.

21. (A) The equator is hot because sunlight arrives nearly overhead, packing its energy onto a small area; near the poles the slanted beam spreads the same energy over more ground, so it warms less. The few thousand kilometres of difference in distance to the Sun (out of about 150 million km) is utterly negligible, so 'closer to the Sun' is wrong. The equator does not get 24-hour daylight; that long-day effect is a polar-summer feature.

Cities and machines are not why the equator is warm — it is the directness of the Sun's rays.

22. (D) Sustained darkness in winter drives temperatures very low, and although polar summer brings long days, the Sun stays low and weak, so the region's climate remains cold overall. The midnight Sun does not make the poles hot; its slanted rays deliver little heat. Darkness does not trap heat to make the region hot — losing incoming sunlight cools it. And the extremes do not 'cancel' into a mild climate; the net result over the year is cold.

23. (B) Same clock time, different exposure, different temperature — this isolates sunlight as the heating agent. If the clock controlled temperature the two spots would read the same, so the clock is not the cause. Shade is cooler because it receives no direct sunlight, not because cold air selectively sinks into it. And the shaded thermometer is not faulty; it correctly reports that an unlit spot is cooler than a sunlit one at the same moment.

24. (D) The atmosphere's many interacting factors amplify tiny differences over time, so distant daily detail is unpredictable, yet the long-run average (the season) stays reliably known — both facts fit together. The Sun does not weaken a year out, so that is not why distant days are unknowable. Climate is very much studiable — indeed it is what lets us predict the season. And thermometers do not stop after a day; instrument failure is not the limit, the atmosphere's sensitivity is.

25. (B) Both effects share one cause: when the Sun is low (winter, or near the poles) its beam hits at a slant and spreads thin, delivering less heat per unit ground. The Earth-Sun distance change across seasons is tiny and is not the reason. Winter days are SHORTER, not longer, so that statement is factually wrong as well. And it is not that winter air physically blocks the light; the weakening comes from the angle of the rays, the same principle that cools the poles.

26. (D) The equator gets concentrated overhead rays year-round (hot); the poles get the same beam smeared over more area at a low angle (cold) — directness of sunlight sets the climate contrast. The two regions do NOT receive equal energy per unit area; that is precisely the difference. Long polar summer days do not overcome the weak slant, so the poles are not warmer. And direct rays carry more heat per area than slanting ones, so the equator-cold / pole-hot claim is exactly backwards.

27. (B) A tropical rainforest is hot all year (small swing) and very wet — City A's 29/27 °C with 2300 mm fits exactly. City B's huge 45-to-8 swing with almost no rain is a hot desert, not a rainforest. City C is freezing (polar), regardless of resembling B's low rainfall. City D is warm but its larger seasonal swing and only moderate 700 mm rainfall fall short of the constant heat and heavy rain a rainforest needs.

28. (A) Constant warmth, very high humidity, abundant year-round rain and a tall canopy are textbook rainforest, and both climbing limbs (for moving among trees) and loud calls

(to communicate where dense leaves block vision) suit it. The data are far too warm and wet for a polar region, and there is no ice to grip. They are far too wet for a desert, and tails do not store water. They are far too warm for a snowfield, and there is no snow to dig. Only the rainforest reading ties every clue together.

29. (B) Fat insulates and stores energy, a compact body conserves heat (low surface-area-to-volume), and white fur hides the animal in snow — all cold-climate features, which is why such an animal would overheat and stand out in a warm green rainforest. They do not help it lose heat; they do the opposite, so it would not stay cool. White fur camouflages against snow, not green leaves, so it would be conspicuous, not hidden. None of these features store water — that is a desert idea, not their purpose here.

30. (D) A compact body has less surface area for its volume, so it radiates heat away more slowly — X is favoured in the cold, while slender Y sheds heat too quickly. Extra surface area loses far more heat than the weak polar Sun could add, so Y does not benefit. Body shape strongly affects heat loss, so the two are not equal. And a compact body exposes LESS relative surface, so X loses heat slower, not faster — the reasoning in that option is inverted.

31. (C) Wide paws serve double duty — distributing weight on soft snow and paddling in water — and the fur and fat together are the insulation that keeps the bear warm in freezing air and sea. The fat's chief role is insulation and energy storage, not floating to ambush prey. The dense fur is for warmth, not for cooling in summer. And swimming is for hunting and moving between ice, not for tree-climbing, which polar bears do not do.

32. (B) The white coat conceals the bear against snow while it stalks seals — its primary job is camouflage, not warming by colour. In fact white reflects rather than absorbs sunlight, so 'white absorbs the most heat' is physically wrong. Standing out to other bears is not why it is white. And being visible to prey would hurt hunting, so making the hunt 'fair' is not an adaptation's purpose — concealment is.

33. (C) By packing together, birds on the inside expose almost no surface to the wind, cutting the group's overall heat loss — the point of huddling. It does not increase exposed surface; that would lose more heat, the opposite of the goal. Huddling on ice has nothing to do with herding fish. And it traps WARM, not cold, air among the birds; trapping cold air would not help them stay warm.

34. (D) Ear surface area governs heat exchange: big thin ears (with flapping) dump excess heat in the warmth, small ears minimise heat loss in the cold — one principle, two opposite climates. The ears are explained by heat regulation here, not primarily by hearing. Large ears SHED heat rather than trap it, so 'large ears trap heat to keep the elephant warm' is wrong. And ear size tracks climate strongly, not body size alone, which is the very point the question makes.

35. (D) Both cases are camouflage: green blends with leaves, white blends with snow — colour is favoured when it matches the local background. Colour here is about concealment, not warming; neither green nor white is chosen to absorb heat. Neither animal glows in the dark, so visibility to its own kind is not the point. And being the brightest to scare predators is the opposite of camouflage, which works by NOT being noticed.

36. (A) Grasping hands and a long tail are for climbing trees and loud calls are for cutting through dense foliage — all rainforest tools useless on bare, treeless, freezing ice, and the monkey also lacks insulation for the cold. The features are not cold-survival traits, so the problem is far more than unfamiliar terrain. They are not burrowing-in-sand traits either — that is a desert idea. And the monkey cannot fly, so 'flying over ocean' misdescribes it entirely.

37. (C) A constant warm, wet climate yields plentiful, diverse resources year-round, which can sustain a great many species — that is why rainforests are so biodiverse. Polar cold does not speed reproduction, and size is not the driver of variety. Rainforests are dense with plants, not nearly bare; abundant plant life is exactly what supports the animals. And biodiversity is set by climate and resources, not by the number of human visitors.

38. (C) A toucan's long light beak lets it pluck fruit from branches too thin to bear its weight, and a polar bear's dense fur is insulation against the cold — each feature matched to its real job. The beak does not store fat, and the fur warms rather than cools the bear, so that pairing fails on both counts. The beak is for feeding, not digging burrows, and the bear's fur camouflages it against snow, not leaves. The beak does not heat the head, and fur is for warmth, not for faster swimming, so the last pairing is wrong throughout.

39. (C) Double fur, fat, a compact body and white colour all conserve heat and hide the animal in snow — the perfect set for cold, icy Habitat N. Thick fur is for cold, not for heavy rain, so Habitat M's warmth and rain do not call for it. White hides an animal in snow, not against green leaves, so it would stand out in M. And insulation is only an advantage where heat must be conserved; in warm M these features would cause overheating, so the habitats are not equal.

40. (D) An adaptation is a heritable feature that suits an animal to its environment over many generations; green that consistently matches rainforest leaves and lowers predation is exactly that. Green colour does not keep a frog warm, and warmth is not the issue in a hot rainforest. The frog is not a desert animal cooling by reflection. And it does not glow to lure prey in darkness — its colour works as daytime camouflage among leaves.

41. (D) An adaptation is inherited and shaped over many generations to fit the climate — successive polar bears with thick fur and fat are the textbook case. One bear feeling warm on a mild day is a momentary experience, not an inherited feature. A monkey learning a

trick in a morning is learned behaviour, not a heritable adaptation. A bird sheltering indoors once is a one-off act, not a feature passed down over long times.

42. (B) Migration tracks the predictable seasonal climate cycle — the bird avoids the long northern winter and returns when conditions there improve, a pattern governed by the season, not one day's weather. It is not the weather of a single departure day that drives the journey. The same cranes return year after year; they do not die out and get replaced. And a bird does not transform into another species; it is the recurring climate pattern, not a weather-triggered transformation, that explains the back-and-forth.

43. (B) Hours jumping between 0 and 9 across six days show large day-to-day variation — a clear picture of changeable weather, not climate. A single week of varying sunshine cannot establish a change in climate, which needs many years. The figures are plainly not constant, so they do not show stability. And the readings differ each day precisely because cloud cover differs, so the Sun did not shine equally regardless of clouds.

44. (B) Freezing temperatures, snow and a very long night signal a polar climate; steady high temperatures, heavy rain and dense forest signal a tropical rainforest. A wide temperature range alone does not make the first a desert — it is the deep cold, snow and long night that matter, and the second is hot and wet, not polar. Precipitation occurs in many climates, so its mere presence does not make both tropical. And a maximum just above zero with deep cold and snow is polar, not a temperate coast, while a hot place with heavy rain and forest is a rainforest, not a desert.

45. (C) A polar region is identified by a long-term cold pattern, whereas one cold day is just weather and occurs in many non-polar places too. Cold days clearly do happen outside polar regions, so the first alternative is false. A single cold day cannot prove a polar climate, so the child is not right. And temperature is central to defining a polar region — it is the persistence of cold, not its absence from the definition, that the child overlooked.

46. (B) With latitude held equal, nearness to the sea is the key remaining factor: the sea evens out coastal temperatures and raises humidity, while inland places, lacking that buffer, have larger swings and drier air. Altitude is not implied by the data and is a separate factor, so it is not the explanation here. Inland places are not closer to the Sun, and that idea is a misconception about sunlight. And the year-long contrast in range and humidity is a lasting climate difference, not an artefact of which single day each was measured.

47. (A) A similar average can mask a real difference in variability: Place 1 is steady (perhaps coastal or rainforest-like) while Place 2 swings wildly (desert-like), and that spread is itself a climate distinction. Equal averages therefore do not mean identical climates. A highly variable place still has a climate — a variable one. And steady temperatures do not mean 'no weather'; daily weather still occurs even when the maxima barely move.

48. (C) Humidity, expressed as a percentage, is the direct measure of water vapour in the air. Rainfall measures water that has fallen as rain, which can be zero even when the air is humid, so it is not a vapour measure. Temperature affects how much vapour air CAN hold but does not state how much is actually present, so a hot reading alone does not give the moisture content. And wind speed measures air movement, not the amount of vapour the air carries.

49. (C) A slender body and large thin ears maximise surface area for losing heat, and short fur offers little insulation — all suited to a hot climate. Large ears shed heat, they do not hold warmth, so they do not suit a polar climate. A slender body LOSES heat faster than a compact one, so it does not conserve heat for the cold — that reasoning is inverted. And body shape and ear size strongly affect heat loss, so the features are not climate-neutral.

50. (A) A smooth curve that repeats every year for a decade is the averaged seasonal pattern — climate; a one-week daily zig-zag with no recurring shape is short-term variation — weather. Being drawn from measurements does not make both climate; weather is also measured. The labels are not assigned by which plot is more detailed, so the long plot is not therefore weather. And although temperature is an element used in both, the difference between the plots is exactly the weather-versus-climate distinction, not a reason to call both weather.

51. (C) The yearly round trip follows the predictable seasonal climate, and birds often leave before the worst weather, cued by reliable signals such as shortening days and dwindling food. It is not driven by one day's random weather, which would come too late and too unreliably. The pattern repeats yearly, so the birds are not permanently settling. And it is a recurring cycle, not a one-time event, while one tree's leaf colour is far too local a signal to coordinate a whole flock's migration.

52. (A) A wider funnel collects more volume but spreads it over a larger base, so the DEPTH is the same as in a narrow gauge — which is precisely why depth is the comparable quantity. The wide gauge does collect more volume, but that does not raise its depth, so it does not read deeper. The narrow gauge does receive less total water over a smaller mouth, so the column is not 'squeezed' to a greater depth. And the gauges CAN be compared — depth makes them comparable despite differing funnel sizes.

53. (D) Scorching dry summers, big day-to-night swings and scant rain are the signature of a hot desert, where dry skies allow rapid heating by day and rapid cooling at night. It is not a rainforest — rainforests are wet with little seasonal swing, whereas this region is dry with huge swings. It is not polar — polar climates are cold year-round, not blazing in summer. And a coastal climate has SMALL swings thanks to the moderating sea, so a 5 °C winter minimum with such large swings does not point to a temperate coast.

54. (B) A fat layer insulates, so it suits the cold but causes overheating in a hot rainforest; grasping limbs are for climbing trees, so they suit a rainforest but are useless on treeless polar ice — each helps in one climate and hinders in the other. They are not both cold-climate aids: a grasping tail does not grip ice, and fat is harmful in the heat. They are not both rainforest aids: fat does not cushion against rain and would cause overheating. And both features clearly depend on climate, so 'climate-independent' is wrong.

55. (A) Climate (the wettest month, an average) is stable and predictable, while the weather of one distant day is not, because the atmosphere's sensitivity amplifies tiny differences as time passes — one principle covers both. The Sun does not behave differently on days versus months in a way that explains this. Rainfall does not 'obey the calendar' with random dates; the predictability gap is about averaging, not calendar magic. And instruments work fine for single days too; the limit is the atmosphere's behaviour, not the instruments.

56. (A) The governing rule is surface-area-to-volume ratio: low ratio (compact) conserves heat for the cold, high ratio (lean, big-eared) sheds heat for the warmth. A compact body loses heat SLOWER, not faster, so that option inverts the logic. Body shape strongly affects heat exchange, so it is not merely about appearance to predators. And these shapes are heat-regulation adaptations, not consequences of how much food is available.

57. (A) Direct, intense equatorial sunlight warms that belt and sustains hot, wet rainforest climates, while weak slanting polar sunlight keeps the poles cold and icy — the three facts combine into this latitude-driven climate pattern. Warm air does not sink (it rises), and ice forms at the cold poles, not the equator, so that option misuses fact 2 and contradicts fact 3. Rising warm air does not leave the equator cold; the equator is hot, and rainforests need warmth, not cold. And sunlight strength is precisely what sets these climate zones, so their positions are not accidental.

58. (D) With no sunlight all night the ground keeps radiating away its stored heat, so the air goes on cooling and is coldest just before sunrise, after the longest stretch of heat loss. The Earth's distance from the Sun does not change between midnight and dawn in any meaningful way, so midnight is not coldest for that reason. The Moon does not draw heat from the ground; cooling is simply the loss of the day's stored heat. And a working thermometer does not drift lower on its own — the falling reading reflects the genuinely cooling air.

59. (C) The temperature peak lags the Sun's peak because, for a while after noon, the ground still absorbs more heat than it radiates away, so the air keeps warming until gains and losses balance in the early afternoon. The Sun is highest at noon, not mid-afternoon, so 'directly overhead in mid-afternoon' is wrong. Heat does not 'travel' from the Sun to the ground over a fixed two hours; sunlight arrives in minutes and the lag is about stored heat. And clouds do not reliably gather at noon; the lag occurs even on cloudless days because of accumulated ground heating.

60. (C) Weather is short-term, climate is the long-term average of weather, and adaptations are inherited features tuned to that long-term climate — all three correctly placed. The first wrong option reverses weather and climate and wrongly ties adaptations to single days. The second wrongly equates weather and climate and confuses adaptations (heritable) with one-day learning. The third is wrong because climate IS measurable (by averaging weather data) and adaptations are deeply linked to climate.